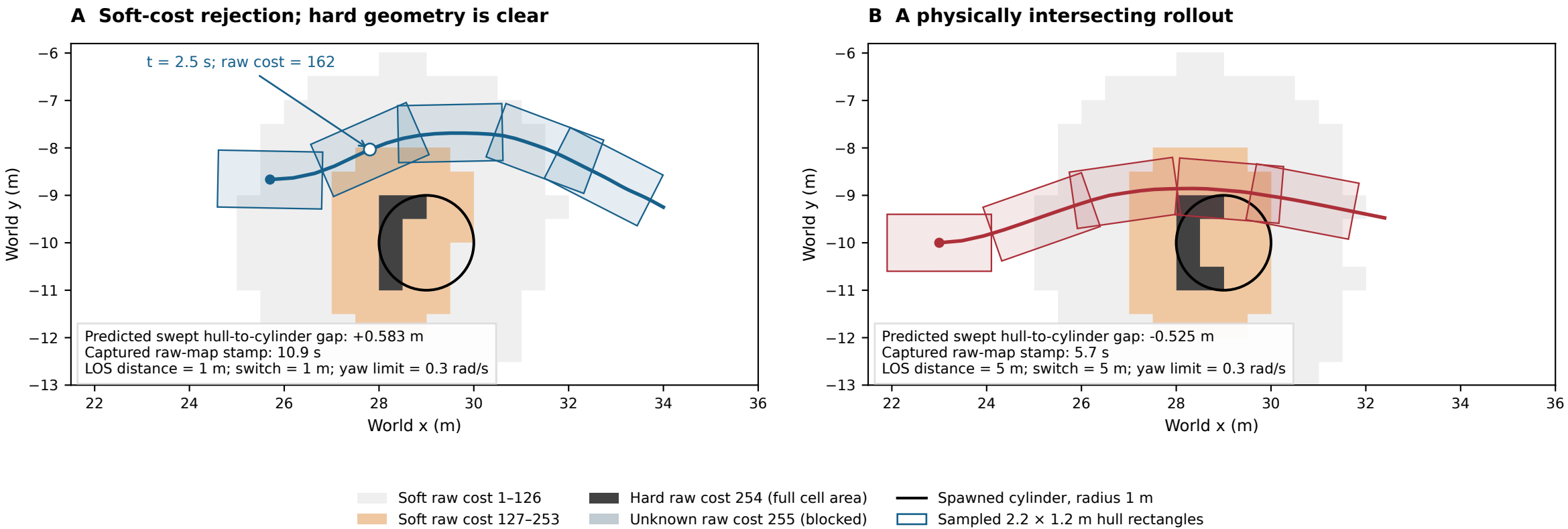
Captured-map diagnosis: soft costs and full-hull collision
A Soft-cost rejection; hard geometry is clear
t = 2.5 s; raw cost = 162
Predicted swept hull-to-cylinder gap: +0.583 m
Captured raw-map stamp: 10.9 s
LOS distance = 1 m; switch = 1 m; yaw limit = 0.3 rad/s
B A physically intersecting rollout
Predicted swept hull-to-cylinder gap: -0.525 m
Captured raw-map stamp: 5.7 s
LOS distance = 5 m; switch = 5 m; yaw limit = 0.3 rad/s
World x (m)
World y (m)
Soft raw cost 1–126
Soft raw cost 127–253
Hard raw cost 254 (full cell area)
Unknown raw cost 255 (blocked)
Spawned cylinder, radius 1 m
Sampled 2.2 × 1.2 m hull rectangles
Measured Gazebo maps; stored offline LOS rollouts. Single-snapshot geometry checks are not closed-loop acceptance results.